HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

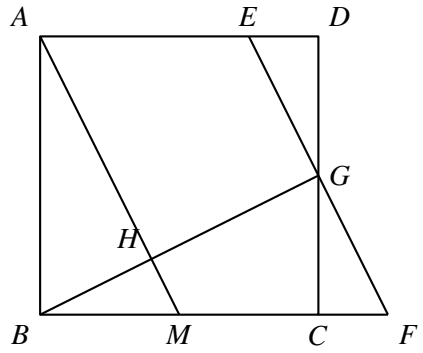
Student No.: 15 Name: _____ Marks: / 30

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, $ABCD$ is a square. Let M be the mid-point of BC. E is a point lying on AD such that $AE : ED = 3 : 1$. F is a point lying on BC produced such that $EF//AM$. CD and EF intersect at the point G while AM and BG intersect at the point H. If the area of $\triangle BMH$ is $4\,\text{cm}^2$, then the area of the trapezium $AEGH$ is



A. $12\,\text{cm}^2$.

B. $33\,\text{cm}^2$.

C. $39\,\text{cm}^2$.

D. $45\,\text{cm}^2$.

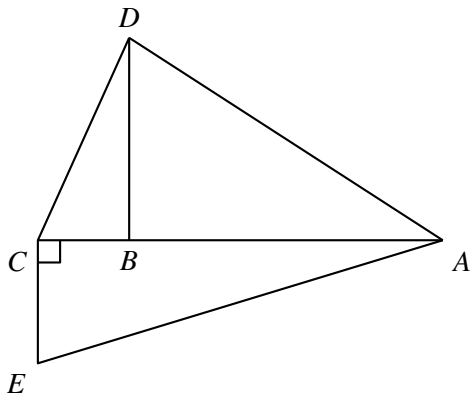
(2 marks) (DSE-2024 PII Q17)

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2. In the figure, ABC is a straight line. It is given that $AD = 37\,\text{cm}$, $BC = 5\,\text{cm}$, $BD = 12\,\text{cm}$, $CD = 13\,\text{cm}$ and $CE = 9\,\text{cm}$. If $\angle ACE = 90°$, find the perimeter of the quadrilateral $ADCE$.



A. 76 cm

B. 90 cm

C. 100 cm

D. 180 cm

(2 marks) (DSE-2024 PII Q18)

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3. If the sum of the interior angles of a regular polygon is 900°, which of the following is/are true?

I. The number of diagonals of the polygon is 7.

II. The number of folds of rotational symmetry of the polygon is 7.

III. The number of axes of reflectional symmetry of the polygon is 7.

A. I only

B. II only

C. I and III only

D. II and III only

(2 marks) (DSE-2024 PII Q20)

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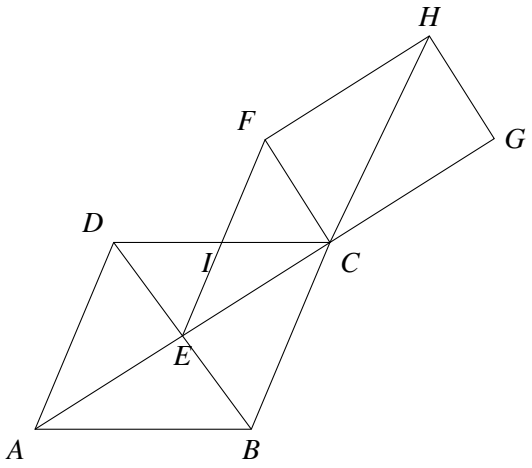
Answers written in the margins will not be marked.

4. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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5. The coordinates of the points A and B are $(-3, 1)$ and $(-7, -5)$ respectively. If P is a point lying on the straight line $x - y + 13 = 0$ such that $AP = PB$, then the y-coordinate of P is

A. -11.

B. -2.

C. 2.

D. 11.

(2 marks) (DSE-2024 PII Q25)

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6. Denote the circle $3x^2 + 3y^2 - 6x + 12y - 4 = 0$ by C. Which of the following are true?

I. The origin lies inside C.

II. The circumference of C is less than 16.

III. The perpendicular distance from the centre of C to the x-axis is 2.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q27)

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7. Two numbers are randomly drawn at the same time from six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is not less than 12.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{7}{15}$

D. $\frac{8}{15}$

(2 marks) (DSE-2024 PII Q28)

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8. If $x = \log_2 y - 2$ and $(\log_2 y)^2 = 5\log_2 y + x - 7$, then $y =$

A. 1.

B. 8.

C. 1 or 3.

D. 3 or 8.

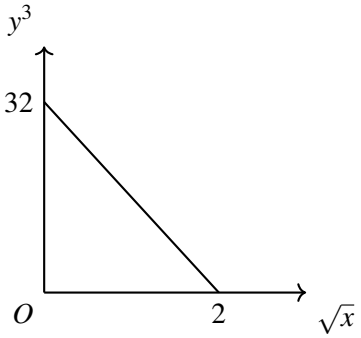
(2 marks) (DSE-2024 PII Q33)

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9. The graph in the figure shows the linear relation between y^3 and $\sqrt{x}$. If $x = 36$, then $y =$



A. -64.

B. -16.

C. -8.

D. -4.

(2 marks) (DSE-2024 PII Q34)

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10. The sum of the first n terms of a sequence is $n(2n+3)$. Which of the following are true?

I. 14 is a term of the sequence.

II. The nth term of the sequence is $4n+1$.

III. The sequence is an arithmetic sequence.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q36)

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11. Consider the following system of inequalities:

$$\begin{cases} x - 2y \leq 1 \\ x + 4y \leq 13 \\ 2x - y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x - 2y + c$ is 22, where (x, y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

(2 marks) (DSE-2024 PII Q37)

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12. For $0° < \theta \leq 360°$, how many roots does the equation $4\cos^2\theta - 3\cos\theta - 1 = 0$ have?

A. 3

B. 4

C. 5

D. 6

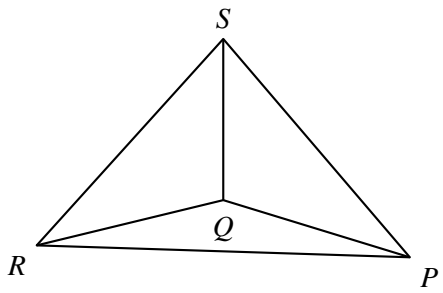
(2 marks) (DSE-2024 PII Q39)

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13. In the figure, the base PQR of the tetrahedron $PQRS$ lies on the horizontal ground. It is given that Q is vertically below S. If $\angle PQR = 90°$, $\angle QPS = 30°$ and $\angle QRS = 45°$, then $\cos\angle PRS =$



A. $\frac{1}{2}$.

B. $\frac{3}{4}$.

C. $\frac{\sqrt{2}}{4}$.

D. $\frac{\sqrt{3}}{6}$.

(2 marks) (DSE-2024 PII Q40)

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14. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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15. A queue is formed by 2 managers and 7 officers. If no managers are next to each other, how many different queues can be formed?

A. 80 640

B. 141 120

C. 282 240

D. 362 880

(2 marks) (DSE-2024 PII Q42)

I can finish _____ questions

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